Pratham Modi

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Founding AI Engineer | Python, Real-Time Voice Pipelines, STT/TTS, LLM Orchestration

Summary

AI Systems Engineer with 2+ years of production experience building low-latency conversational voice pipelines, real-time audio streaming infrastructure, and LLM-driven agentic architectures. Engineered real-time WebSocket communication infrastructure and Kafka event-streaming pipelines handling 75K+ daily events with sub-50ms latency at C3iHub (IIT Kanpur), integrated STT/LLM/TTS tool-calling pipelines, and optimized state tracking across Redis and PostgreSQL. Strong engineering foundation in Python, FastAPI, WebSockets, WebRTC, and prompt engineering.

Technical Skills

Languages & Frameworks: Python (Asyncio, Profiling), React, Node.js, TypeScript, SQL, Go, Java, C++
AI, Voice & Real-Time: Artificial Intelligence, Machine Learning, Real-Time Audio Streaming, WebSockets, WebRTC, STT/TTS Integration, LLM Orchestration, Prompt Design, RAG Pipelines, Vector Databases
Backend Systems: Python (FastAPI), RESTful APIs, Microservices, Distributed Systems, Event-Driven Architecture
Data & Caching: Redis, Redis Streams, PostgreSQL, Apache Kafka, Conversation State Stores, Compound Indexing
Cloud & Observability: AWS (EKS, EC2, S3), Docker, Kubernetes, CI/CD, OpenTelemetry, Prometheus, Grafana

Experience

C3iHub, IIT Kanpur Jul 2025 – Present
Software Development Engineer (Voice AI & Infrastructure) Bengaluru / Kanpur, India

- Architected core platform backend and real-time streaming infrastructure serving 50K+ active users across distributed microservices, ensuring 99.9% uptime via automated health reconciliation.
- Built real-time interactive audio and message streaming modules using WebSockets and async Python workers, achieving sub-100ms transport latencies under concurrent user sessions.
- Engineered event-driven streaming pipeline handling 75K+ daily events with sub-50ms latency using Apache Kafka and Redis Streams – implemented consumer-group partition balancing and idempotent consumer processing.
- Optimized conversation state queries across PostgreSQL and Redis, slashing p99 state retrieval latency from 450ms to 135ms.
- Automated container build, vulnerability scanning, and deployment pipelines on AWS using GitHub Actions, cutting staging deploy cycles by 55%.

Playpower Labs Jan 2024 – Jun 2025
Software Development Engineer (Backend & AI Services) Bengaluru, India

- Built high-performance async microservices in Python (FastAPI) and Node.js handling 500+ persistent concurrent WebSocket connections with p95 response latencies under 85ms.
- Orchestrated STT/LLM/TTS conversational pipelines with prompt engineering and structured tool execution for automated voice interactions.
- Implemented distributed caching and query optimization across PostgreSQL and Redis, improving database throughput by 40% under burst traffic workloads.
- Integrated OpenTelemetry distributed tracing across 15+ backend services, instrumenting unified spans that reduced cross-service incident resolution time by 35%.

Projects

Real-Time Voice Agent Orchestrator | *Python, WebSockets, WebRTC, Redis, PostgreSQL, Docker, AWS*

- Engineered real-time streaming voice agent connecting audio ingestion to STT, LLM reasoning, and TTS synthesis with voice activity detection and natural turn-taking controls.
- Implemented session context and persistent state management using Redis Streams and PostgreSQL, enabling seamless multi-turn conversation recovery.

Education

Pandit Deendayal Energy University 2021 – 2025
Bachelor of Technology in Computer Science and Engineering CGPA: 9.20 / 10.0